# Yangmengfei (Murphy) Xu

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## EDUCATION

**Doctor of Philosophy** (Engineering and IT), The University of Melbourne, expected thesis submission: late 2026

- Rowden White Scholarship (awarded to the highest-ranked applicant in the department; one recipient per year)
- Melbourne Research Scholarship

**Master of Philosophy** (Engineering and IT), The University of Melbourne, 2022

- Melbourne Research Scholarship (Fee Offset)

**Master of Engineering** (Mechatronics Engineering), The University of Melbourne, 2018

**Bachelor of Engineering** (Electrical & Electronic Engineering), The University of Liverpool, 2016

**Bachelor of Engineering** (Electrical & Electronic Engineering), Xi'an Jiaotong-Liverpool University, 2016

## TEACHING EXPERIENCE

**Course Tutor and Project Supervisor,** The University of Melbourne, 2021-Present

- Deliver tutorials and laboratory workshops across engineering and robotics subjects; prepare teaching materials and assessments, mark assignments and examinations, and provide structured feedback to students.
- Co-supervised more than 30 capstone and research projects, providing guidance on project planning, experimental design, implementation, and weekly progress, with selected projects leading to publishable research outcomes.

## RESEARCH EXPERIENCE

**Control and Planning Design for Autonomous Robots in Mining Engineering,** Researcher

- Developed robust control and planning methods for autonomous and multi-robot systems, including Spatial Iterative Learning Control, Active Disturbance Rejection Control for non-holonomic robots, and geometry-based distributed connectivity maintenance for visual-sensing agents.
- Demonstrated disturbance rejection in harsh environments (Unmanned Systems, 2023), connectivity preservation under uncertainty (Unmanned Systems, 2024), and spatial learning control under input saturation and variable task duration (IEEE/ASME Transactions on Mechatronics, 2026).

**Human Motor Adaptation in Rehabilitation Robotics,** Researcher

- Designed and conducted human experiments to induce implicit motor adaptation through energy-based cost shaping using a manipulandum platform; developed the Indirect Shaping Control framework for movement retraining.
- Demonstrated indirect and implicit motor adaptation through robot-generated force fields (IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2021), and established a quantitative link between human motor cost and movement adaptation (IFAC, 2026).

**Autonomous Community-Sharing Service Robot,** Project Leader

- Designed and developed an autonomous community-sharing robot to foster social connection and sustainability through item exchange and collective memory creation.
- Demonstrated how robotics can support community belonging and environmental responsibility; the prototype was presented at the HRI 2025 Student Design Competition and was awarded the HRI 2025 Sustainability Recognition.

**Epistemic Logic Modeling of Human Theory of Mind,** Project Leader

- Developed the Predictive Justified Perspective (PJP) model for representing and predicting higher-order beliefs in dynamic multi-agent environments, with applications to human–robot interaction and epistemic planning.
- Investigated bounded human reasoning through pilot user studies (CHI EA, 2025) and predictive belief updates for dynamic epistemic planning using PJP (SoCS, 2026).

**PyPose Open-Source Robotics Library,** Developer

- Contributed core capabilities to PyPose, an open-source robotics library with 1.6k+ GitHub stars, focusing on point-cloud processing and registration; co-authored the PyPose v0.6 paper presented at the IROS 2023 Workshop.

## INDUSTRIAL EXPERIENCE

**Research Engineer,** Boton Technology & The University of Melbourne**,** 2021-2023

- Led an industry–university collaboration to develop autonomous cleaning robots for material-handling systems in the mining sector.
- Developed system architecture, autonomous navigation, computer vision, and AI decision-making modules for large-scale industrial deployment.

## ADDITIONAL ACTIVITIES

**Ausdroid AI & Robotics Group,** Founder and President, 2017-Present

**Suzhou Ocean Paradise Autistic Children's Rehabilitation Center,** Volunteer, 2012-2013

## PUBLICATIONS

**Xu, Y.,** Tan, Y., & Oetomo, D. (2026). Spatial Iterative Learning Control for Nonholonomic Robotics With Input Saturation. *IEEE/ASME Transactions on Mechatronics*, early access: https://doi.org/10.1109/TMECH.2026.3721852.

**Xu, Y.,** Crocher, V., Fong, J., Tan, Y., & Oetomo, D. (2026, August). Motor Cost Re-Optimization in Indirect Human Movement Pattern Adaptation. Presented at the *23rd IFAC World Congress*, Busan, South Korea.

Hu, G., Li, W., & **Xu, Y**. (2026, August). Beyond Static Assumptions: the Predictive Justified Perspective Model for Epistemic Planning. In *Proceedings of the International Symposium on Combinatorial Search* (Vol. 19, No. 1, pp. 65-73).

Li, W., Zhang, C., Li, W., Hu, G., & **Xu, Y.** (2025, April). Modeling Higher-order Human Beliefs Using the Justified Perspective Model. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (pp. 1-8).

**Xu, Y.,** Tian, S., Wang, G., & Tang, B. (2025, March). From Isolation to Connection: Community Service Robots for Social Cohesion and Sustainability. In *ACM/IEEE International Conference on Human-Robot Interaction (HRI)* (pp. 1953-1956).

Li, X., Fu, J., Liu, M., **Xu, Y.,** Tan, Y., Xin, Y., ... & Oetomo, D. (2024). A Geometry-Based Distributed Connectivity Maintenance Algorithm for Discrete-time Multi-Agent Systems with Visual Sensing Constraints. *Unmanned Systems*, *12*(02), 261-275.

Liu, M., **Xu, Y.,** Lin, X., Tan, Y., Pu, Y., Li, W., & Oetomo, D. (2024). On active disturbance rejection control for unmanned tracked ground vehicles with nonsmooth disturbances. *Unmanned Systems*, *12*(06), 1023-1037.

Zhan, Z., Li, X., Li, Q., He, H., Pandey, A., Xiao, H., **Xu, Y.,** ... & Wang, C. (2023). PyPose v0.6: The imperative programming interface for robotics. In 2023 *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) Workshop*.

**Xu, Y.,** Crocher, V., Fong, J., Tan, Y., & Oetomo, D. (2021). Inducing human motor adaptation without explicit error feedback: a motor cost approach. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, *29*, 1403-1412.

Mohammadi, A., **Xu, Y.,** Tan, Y., Choong, P., & Oetomo, D. (2019). Magnetic-based soft tactile sensors with deformable continuous force transfer medium for resolving contact locations in robotic grasping and manipulation. *Sensors*, *19*(22), 4925.